

ETHICAL HACKING TASK 02

Network Scanning & Service Enumeration Report

Target: localhost (127.0.0.1)

System: Personal Windows 11 Machine

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Date: June 27, 2026

White Band Associates Internship

Part A: Nmap Installation

Tool: Nmap 7.99

Platform: i686-pc-windows-windows

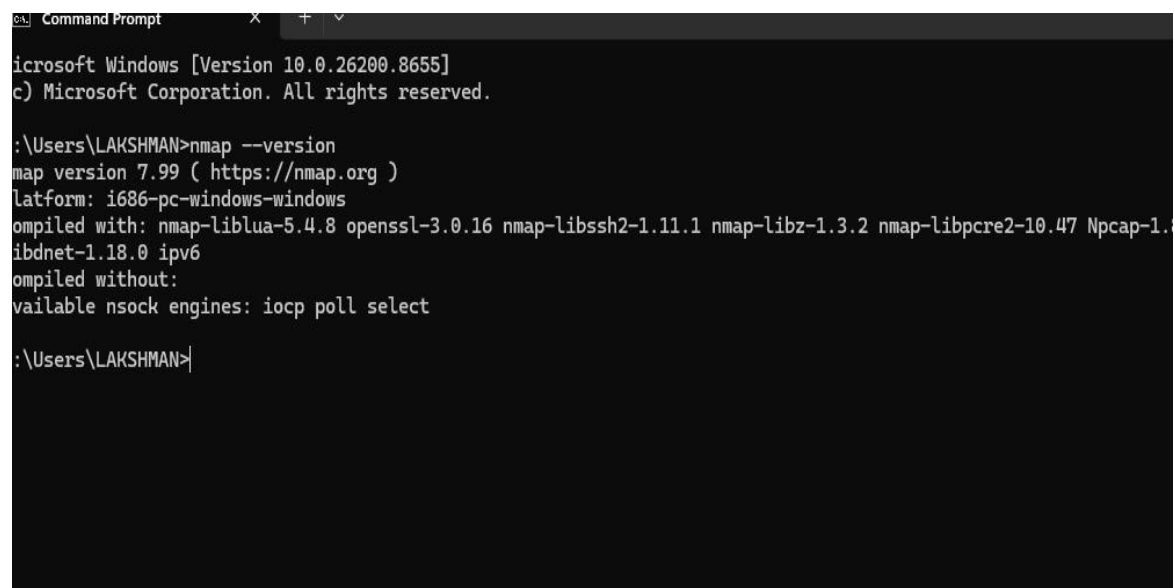
Npcap Version: 1.87

Installation Source: <https://nmap.org/download.html>

Verification command used:

```
nmap --version
```

Screenshot

A screenshot of a Windows Command Prompt window. The title bar reads 'Command Prompt'. The window content shows the output of the 'nmap --version' command. The text is as follows:
Microsoft Windows [Version 10.0.26200.8655]
c) Microsoft Corporation. All rights reserved.

C:\Users\LAKSHMAN>nmap --version
nmap version 7.99 (https://nmap.org)
Platform: i686-pc-windows-windows
Compiled with: nmap-liblua-5.4.8 openssl-3.0.16 nmap-libssh2-1.11.1 nmap-libz-1.3.2 nmap-libpcap-1.0.4 nmap-libnmap-1.0.0 nmap-libnet-1.1.8 nmap-libidn2-2.3.0 nmap-libc-1.0.0 nmap-libbsdnet-1.18.0 ipv6
Compiled without:
Available nsock engines: iocp poll select

C:\Users\LAKSHMAN>

Screenshot 1: nmap --version — confirms Nmap 7.99 installed successfully on Windows 11

Part B: Basic Localhost Scan

Command used:

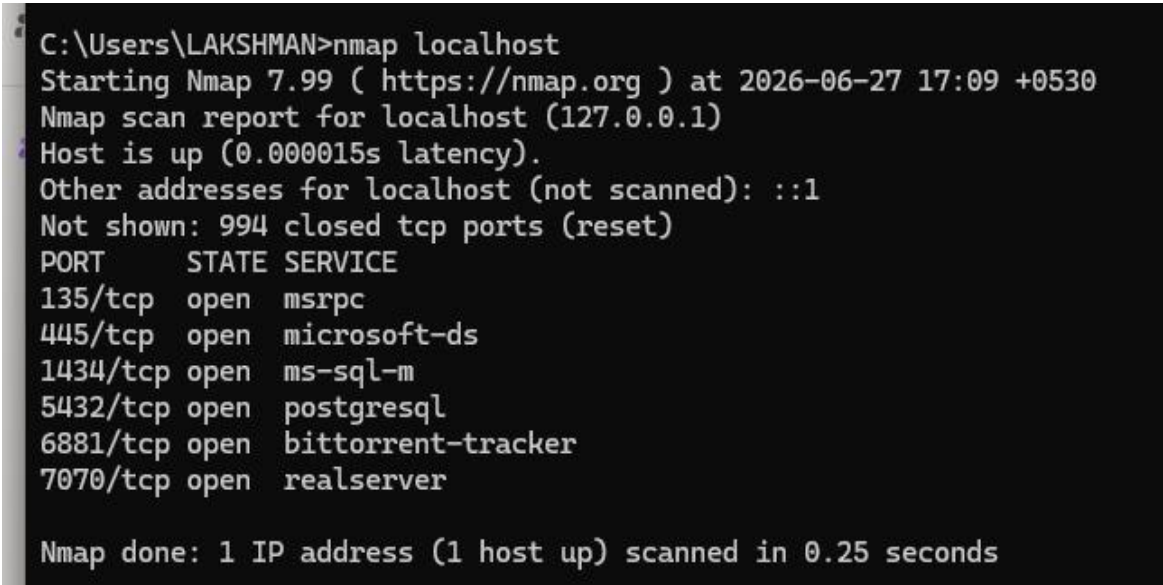
```
nmap localhost
```

Field	Value
Target	localhost (127.0.0.1)
Host Status	Up (0.000015s latency)
Total Ports Scanned	1000
Closed Ports	994
Open Ports	6
Scan Time	0.25 seconds

Open Ports Found

Port	State	Service
135/tcp	open	msrpc
445/tcp	open	microsoft-ds
1434/tcp	open	ms-sql-m
5432/tcp	open	postgresql
6881/tcp	open	bittorrent-tracker
7070/tcp	open	realserver

Screenshot



Screenshot 2: nmap localhost — 6 open ports found including SQL Server, PostgreSQL and BitTorrent

Part C: Service Version Detection

Command used:

```
nmap -sV localhost
```

Port	Service	Version
135/tcp	msrpc	Microsoft Windows RPC
445/tcp	microsoft-ds	Not fully detected
1434/tcp	ms-sql-s	Microsoft SQL Server 2022 16.00.4210
5432/tcp	postgresql	PostgreSQL DB
6881/tcp	tcpwrapped	Not detected
7070/tcp	ssl/realserver	Not fully detected

- Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows
- MS SQL Server 2022 version 16.00.4210 confirmed on port 1434
- Service detection time: 39.15 seconds

Screenshot

```
C:\Users\LAKSHMAN>nmap -sV localhost
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-27 17:10 +0530
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000015s latency).
Other addresses for localhost (not scanned): ::1
Not shown: 994 closed tcp ports (reset)
PORT      STATE SERVICE          VERSION
135/tcp    open  msrcpc           Microsoft Windows RPC
445/tcp    open  microsoft-ds?
1434/tcp   open  ms-sql-s         Microsoft SQL Server 2022 16.00.4210
5432/tcp   open  postgresql       PostgreSQL DB
6881/tcp   open  tcpwrapped
7070/tcp   open  ssl/realserver?
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 39.15 seconds
```

Screenshot 3: `nmap -sV localhost` — MS SQL Server 2022 and PostgreSQL versions identified

Part D: Operating System Detection

Command used:

```
nmap -O localhost
```

Field	Value
OS Detected	Yes
Device Type	General Purpose
Running	Microsoft Windows 11
OS CPE	cpe:/o:microsoft:windows_11
OS Details	Microsoft Windows 11 24H2 – 25H2
Network Distance	0 hops

Why is OS Detection Useful?

- Identifies OS-specific vulnerabilities and unpatched CVEs relevant to the target.
- Allows penetration testers to select appropriate exploits for the detected OS version.
- Reveals the exact Windows build — older builds may have known critical vulnerabilities.
- Helps map expected default services and configurations running on the system.

Screenshot

```
C:\Users\LAKSHMAN>nmap -O localhost
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-27 17:10 +0530
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000042s latency).
Other addresses for localhost (not scanned): ::1
Not shown: 994 closed tcp ports (reset)
PORT      STATE SERVICE
135/tcp    open  msrpc
445/tcp    open  microsoft-ds
1434/tcp   open  ms-sql-m
5432/tcp   open  postgresql
6881/tcp   open  bittorrent-tracker
7070/tcp   open  realserver
Device type: general purpose
Running: Microsoft Windows 11
OS CPE: cpe:/o:microsoft:windows_11
OS details: Microsoft Windows 11 24H2 – 25H2
Network Distance: 0 hops

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 1.28 seconds
```

Screenshot 4: `nmap -O localhost` — OS detected as Microsoft Windows 11 24H2–25H2

Part E: Common Port Research

Port	Protocol	Purpose	Common Security Risks
20/21	FTP	File transfer between client and server	Plaintext transmission, anonymous login, brute force
22	SSH	Encrypted remote login and command execution	Brute force on weak passwords, outdated SSH versions
23	Telnet	Unencrypted remote login	Everything in plaintext including passwords — highly insecure
25	SMTP	Sends emails between mail servers	Open relay abuse for spam, email spoofing without SPF/DKIM
53	DNS	Resolves domain names to IP addresses	DNS poisoning, DNS amplification DDoS attacks
80	HTTP	Serves unencrypted web pages	MITM attacks, credential sniffing, no encryption
110	POP3	Downloads emails from server to client	Plaintext credentials unless SSL/TLS used
143	IMAP	Access emails stored on mail server	Plaintext unless IMAPS (993) — vulnerable to sniffing
443	HTTPS	Encrypted web traffic over TLS/SSL	Misconfigured SSL, outdated ciphers, certificate spoofing
445	SMB	Windows file and printer sharing	EternalBlue/WannaCry ransomware — critical risk if exposed
3389	RDP	Remote graphical desktop access to Windows	Brute force, BlueKeep CVE-2019-0708 — never expose to internet

Part F: Scan Analysis

1. Which services are currently running?

Six services were found running on localhost:

- Port 135 — Microsoft RPC: core Windows inter-process communication requirement
- Port 445 — SMB: Windows file and printer sharing
- Port 1434 — Microsoft SQL Server 2022: active database server (version 16.00.4210)
- Port 5432 — PostgreSQL: open-source database server, likely used for development
- Port 6881 — BitTorrent tracker: peer-to-peer file sharing service (third-party)
- Port 7070 — RealServer: media streaming service (third-party)

2. Are all open ports necessary?

Port 135 (RPC) is a Windows core requirement. Port 445 (SMB) is needed only if file sharing is actively used. Ports 1434 and 5432 suggest the machine is used for database development or runs applications requiring SQL Server and PostgreSQL. However, port 6881 (BitTorrent) and port 7070 (RealServer) are third-party services with no standard system purpose and should be reviewed for necessity.

3. Which services could become security risks if misconfigured?

- Port 445 (SMB) — EternalBlue/WannaCry attack vector, highly dangerous if exposed to a network
- Port 1434 (MS SQL Server) — risk of SQL injection and unauthorized database access if improperly secured
- Port 5432 (PostgreSQL) — database exposure and data theft risk if default credentials are in use
- Port 6881 (BitTorrent) — potential malware distribution vector and policy violation risk
- Port 7070 (RealServer) — outdated streaming service, may have unpatched vulnerabilities

4. Which port would you disable if not required?

Port 6881 (BitTorrent tracker) — this is not a standard system service and poses both security and organizational policy risks. It should be disabled immediately if not intentionally and explicitly required. Port 7070 (RealServer) should also be disabled if media streaming is not actively in use.

Part G: Scan Report & Conclusion

Scan Summary

Field	Details
Scan Date	June 27, 2026
Target System	localhost (127.0.0.1)
Operating System	Microsoft Windows 11 24H2 – 25H2
Nmap Version	7.99
Commands Used	nmap --version nmap localhost nmap -sV localhost nmap -O localhost
Total Open Ports	6 (135, 445, 1434, 5432, 6881, 7070)
Running Services	Windows RPC, SMB, MS SQL Server 2022, PostgreSQL, BitTorrent, RealServer

Observations

- 6 open ports found — more than a default Windows installation would typically expose.
- Two database servers running: MS SQL Server 2022 (1434) and PostgreSQL (5432).
- BitTorrent tracker (6881) and RealServer (7070) are non-standard services — likely third-party installs.
- OS correctly identified as Windows 11 24H2–25H2 by Nmap fingerprinting.
- No HTTP/HTTPS server running — machine is not acting as a web server.

Recommendations

- Disable port 6881 (BitTorrent) immediately if not intentionally required — security and policy risk.
- Review and disable port 7070 (RealServer) if media streaming is not actively needed.
- Restrict port 445 (SMB) via Windows Firewall if file sharing is not required on this machine.
- Ensure MS SQL Server 2022 and PostgreSQL are secured with strong credentials and not exposed to the network.
- Keep Windows 11 fully updated to ensure all SMB and RPC patches are applied.

Conclusion

The network scanning phase on my local Windows 11 machine using Nmap 7.99 revealed six open ports — significantly more than a standard Windows installation. Beyond the default Windows services (RPC on port 135 and SMB on port 445), the scan identified two active database servers: Microsoft SQL Server 2022 (port 1434) and PostgreSQL (port 5432), indicating the machine is used for software development or application hosting.

More notably, port 6881 (BitTorrent tracker) and port 7070 (RealServer) were found open — both are third-party services with no standard system purpose. BitTorrent in particular poses

both security risks (potential malware vector) and organizational policy concerns and should be disabled unless explicitly required.

From a security standpoint, SMB (port 445) remains the most critical risk — it was the attack vector for the WannaCry ransomware attack. Both database ports must be properly secured with strong credentials and restricted from external access. This exercise showed how a simple Nmap scan can quickly reveal a machine's complete service footprint, including unexpected services that might otherwise go unnoticed.

Disclaimer: All scans were performed exclusively on localhost (127.0.0.1) — my own personal Windows 11 machine. No external systems, public IPs, or unauthorized machines were scanned at any point. This report is submitted as part of the White Band Associates Ethical Hacking Internship Task 02.